Nonlocal Self-Similarity-Based Hyperspectral Remote Sensing Image Denoising With 3-D Convolutional Neural Network

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***Abstract*— Recently, deep-learning-based denoising methods for hyperspectral images (HSIs) have been comprehensively studied and achieved impressive performance because they can effectively extract complex and nonlinear image features. Compared with deep-learning-based methods, the nonlocal similarity-based denoising methods are more suitable for images containing edges or regular textures. We propose a powerful HSI denoising method, termed non-local 3-D convolutional neural network (NL-3DCNN), combining traditional machine learning and deep learning techniques. NL-3DCNN exploits the high spectral correlation of an HSI using subspace representation, and the corresponding representation coefficients are termed eigenimages. The high spatial correlation in eigenimages is exploited by grouping nonlocal similar patches, which are denoised by a 3-D convolutional neural network. The numerical and graphical denoising results of the simulated and real data show that the proposed method is superior to the state-of-the-art methods.**

***Index Terms*— 3DCNN, deep learning, denoising, hyperspectral image restoration, nonlocal patch (cube).**

I. Introduction

HYPERSPECTRAL images (HSIs) obtained by imaging spectrometer have rich spectral information and play an essential role in Earth observation, including environmental protection, mineral exploration, and forest monitoring [1], [2]. The rich spectral information in HSI provides an effective solution for the refined recognition of materials. However, due to the complexity of imaging, there are often various noises in hyperspectral images, such as Gaussian noise, Poisson noise, and stripe noise. The noise in HSI has a particular impact on the improvement of the accuracy for many application fields, including unmixing [3]–[5], classification [6], and target detection [7], [8]. Therefore, HSI denoising is an important image preprocessing step in the process of information extraction [9]. In this article, we focus on the additive Gaussian noise in HSI.

To solve this problem, the existing approaches can be mainly divided into three categories: filter-based methods, optimization-based methods, and deep-learning-based methods.

A. Filter-Based Methods

The filtering-based methods include the spatial filtering methods and transform domain methods, which reduce the noise in the transform domain. Specifically, the methods based on the spatial domain are given as follows: a hybrid spatial–spectral noise reduction (HSSNR) method is proposed in [10], where the dissimilarity between the spatial and spectral dimension is explored, that is, the noise is first removed from the spatial dimension, and then removed in the spectral domain, separably. In addition, a generalized multidimensional Wiener filter for denoising is adapted to HSI [11], which considers multidimensional data as a third-order tensor. Later, a Gabor filter-based method is proposed for exploiting the sparsity and the orientation of the stripes in HSIs [12]. Meanwhile, various methods were proposed based on the transform domain filtering [13], [14]. Specifically, a hyperspectral denoising method was proposed using the principal component analysis (PCA) in [13], where the 2-D bivariate wavelet thresholding method and a 1-D dual-tree complex wavelet transform method are introduced to reduce the noise. The method proposed in [14] is a grouping and collaborative filtering paradigm and is widely used as a comparison method in the study of HSI denoising problem.

B. Optimization-Based Methods

With the development of the denoising methods, various optimization-based methods were proposed for exploring the self-similarity, low-rankness, and spatial–spectral information in HSIs [8], [15], [16]. Specifically, self-similarity, which is

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almost or indeed similar to a part of it, is an essential feature of images and has been studied extensively in the state-of-the-art denoising methods [17]–[22]. Meanwhile, as high-dimensional images tend to gather in the dimensional space, there is a strong correlation among the spectral bands of HSIs. What is more, spatial low-rank structure also exists in HSIs due to the spatial aggregation of materials [5], [23], [24]. Therefore, a large number of methods based on subspace learning and low-rank matrix decomposition have been proposed for exploring low-rank representation in HSIs [8], [17], [25]–[29]. However, all the methods mentioned above are matrix-based methods, which transform the original image into a 2-D matrix for processing and damage the inherent spatial information in the image. Considering that the HSI can be defined as a third tensor naturally, which can keep the spatial–spectral information in the image well; hence, tensor-based methods have been studied in recent years and gradually became a research hotspot [30]–[34].

C. Deep-Learning-Based Methods

In the past few years, deep-learning-based methods have achieved outstanding results in the field of natural image processing [35] and have been introduced to the remote sensing community successfully [36]. Correspondingly, a large number of neural-network-based methods have been proposed for HSI denoising [37]–[50]. Most of these are supervised-based approaches: Chang *et al.* [51] proposed a convolutional neural network (CNN)-based HSI denoising method (HSI-DeNet), where the dilated convolution and the multichannel filters were used to aggregate multiscale contextual information of the image. Later, a single-band-based HSI deep CNN (HSI-DCNN) method was proposed in [52], considering the spatial–spectral information in HSIs. Specifically, a single band and its adjacent bands were used simultaneously to explore the spectral correction. Meanwhile, the multiscale feature extraction technique and multilevel feature representation were used to enhance the feature expression. In addition, a single-model-based CNN HSI denoising method (HSI-SDeCNN) is proposed in [53], in which a noise-level map is used to tradeoff the denoising results and the detail information, and the spectral information is composed by a single band and its adjacent bands. In that case, the HSI-SDeCNN method can deal with all noise levels with only one trained model. Furthermore, Wei *et al.* [46] proposed a quasi-recurrent neural network (QRNN3D), which can explore spatio-spectral correlation and global correlation simultaneously.

Although deep-learning-based methods achieved impressive performance in HSI denoising, they still do not fully dig out the spectral correction. Within the scope of our knowledge, there is not yet a deep learning denoising network using all the bands of an HSI as input because of the large number of bands, which brings great difficulties to feature extraction (in the sense of the number of network parameters, training time, and computing power). Meanwhile, a network trained using all the bands of HSIs usually could not be applied to images with a different number of bands. HSIs acquired by different sensors tend to have a different number of bands. To alleviate this problem, two strategies can be used: on one hand, single-band denoising strategy is adopted, and neighboring bands are considered to use spectral information, such as a single band together with its adjacent K bands were considered in HSID-CNN [52] and HSI-SDeCNN [53]. The single-band-based HSI denoising methods have strong robustness and can be applied to images with different bands. On the other hand, another strategy is to train a subset of the band. For example, the Interdisciplinary Computational Vision Laboratory (ICVL) dataset [54] was acquired by Speicm PS Kappa DX4 hyperspectral camera, which contains 519 spectral bands, whose ten bands ranging from $400 to 500$ nm were used to train the network in [51]. Therefore, the trained network in [51] can only denoise ten bands in the real HSI at a time and cannot take full advantage of spectral correlation in all the bands. There is still room to improve the deep-learning-based HSI denoising performance by fully exploiting correlation existing in all the bands. Meanwhile, the deep-learning-based methods have strong data fitting ability, but poor interpretability. Hence, several hybrid methods are proposed, which can enhance the interpretability of networks [55], [56].

D. Motivation and Contribution

To fully exploit correlation existing in all the bands and not to feed all bands into a network, we propose a novel HSI denoising method, termed non-local 3-D convolutional neural network (NL-3DCNN).

1) The proposed NL-3DCNN method takes full advantage of HSIs' structure: spectral low-rankness, and spatial nonlocal self-similarity. Spectral vectors of an HSI are represented in a prelearned low-dimensional subspace, and the corresponding representation coefficients are termed eigenimages. Using subspace representation of HSIs enables us to feed low-dimensional patches of eigenimages (instead of original images with all the bands or a subset of bands) into a deep network. This strategy improves the generalization ability of our denoising network.
2) The nonlocal self-similarity of HSIs is exploited using a 3-D CNN (instead of low-rank or sparse representations used in current nonlocal self-similarity-based HSI denoisers). To the best of our knowledge, it is the innovative HSI denoiser that combines the traditional machine learning (i.e., nonlocal patch-based filtering) and deep learning techniques (i.e., 3DCNN).

The rest of this article is organized as follows. Section II introduces the mathematics definitions and subspace representation of HSI. Section III details the proposed grouped similar patches denoising with the 3-D CNN method. Sections IV and V illustrate the experimental results for the synthetic data and real dataset, respectively. Section VI gives the conclusion of this article.

II. Problem Formulation

A. Notation and Definitions

In this section, we first introduce the notions and definitions used in this article. The matrix is in bold capital letters, and the

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tensor is identified using Euler-script letters—e.g., $\mathbf{X} \in \mathbb{R}^{I_1 \times I_2}$ is a 2-D matrix. The $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ is a third-order tensor having three dimensions. Two definitions of tensors are given as follows.

Definition 1: The dimension of a tensor is called the mode: hence, an HSI can be naturally represented as a third-order tensor with three modes—e.g., $\mathcal{T} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, which consists of three modes, in which the I_1 mode and I_2 mode represent the spatial domain and the I_3 mode stands for the spectral domain.

Definition 2: The mode-3 product of a third-order tensor $\mathcal{Q} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ by a matrix $\mathbf{X} \in \mathbb{R}^{J_1 \times I_3}$ is a tensor $\mathcal{G} \in \mathbb{R}^{I_1 \times I_2 \times J_1}$, denoted as

$$\mathcal{G} = \mathcal{Q} \times_n \mathbf{X}$$

where each element of $\mathcal{G}$ is defined as the sum of products of the corresponding entries in $\mathcal{Q}$ and $\mathbf{X}$

$$\mathcal{G}(i_1, i_2, j_1) = \sum_{i_3} \mathcal{Q}(i_1, i_2, i_3) \cdot \mathbf{X}(j_1, i_3).$$

B. Subspace Representation of HSI

The hyperspectral image cube can be expressed as a third-order tensor data, i.e., $\mathcal{Y} \in \mathbb{R}^{n_{\text{row}} \times n_{\text{col}} \times L}$, with the size of $n_{\text{row}} \times n_{\text{col}}$ and L spectral bands. Assuming that the noise in the image follows a Gaussian distribution, the observed hyperspectral image can be defined as follows:

$$\mathcal{Y} = \mathcal{A} + \mathcal{N} \tag{1}$$

where $\mathcal{A} \in \mathbb{R}^{n_{\text{row}} \times n_{\text{col}} \times L}$ and $\mathcal{N} \in \mathbb{R}^{n_{\text{row}} \times n_{\text{col}} \times L}$ denote the clean hyperspectral cube and the white Gaussian noise cube, respectively.

The spectral vectors of a hyperspectral image live in a low-dimensional subspace because of the strong correlation between spectral bands [17], [57]. The subspace representation theory has been widely used in hyperspectral images and has shown promising results [9]. Based on the subspace theory, $\mathcal{A}$ can be expressed as follows:

$$\mathcal{A} = \mathcal{X} \times_3 \mathbf{C} \tag{2}$$

where $\mathcal{X} \in \mathbb{R}^{n_{\text{row}} \times n_{\text{col}} \times p}$ is the coefficient cube, also called the eigenimages. $\mathbf{C} \in \mathbb{R}^{L \times p}$ is the orthogonal basis matrix. Furthermore, the dimension of subspace, p, is much smaller than L. Hence, HSI with formulation (2) can be rewritten as follows:

$$\mathcal{Y} = \mathcal{X} \times_3 \mathbf{C} + \mathcal{N}. \tag{3}$$

C. Eigenimage-Based Denoising

Assume that the noise in the observed HSI $\mathcal{Y}$ is the zero-mean Gaussian noise, then $\mathbf{C}$ can be learned from the observed image by performing singular value decomposition of $\mathcal{Y}_{(3)}$ (see reference [17] for more details). Hence, the eigenimage-based denoising method can be defined as follows:

$$\widehat{\mathcal{X}} = \arg\min_{\mathcal{X}} \ \frac{1}{2}\|\mathcal{Y} - \mathcal{X} \times_3 \mathbf{C}\|_F^2 + \lambda\phi(\mathcal{X}) \tag{4}$$

$$= \arg\min_{\mathcal{X}} \ \frac{1}{2}\left\|\mathcal{Y} \times_3 \mathbf{C}^T - \mathcal{X}\right\|_F^2 + \lambda\phi(\mathcal{X}) \tag{5}$$

where $\|\mathcal{X}\|_F^2$ denotes the Frobenius norm, which returns the square root of the sum of the absolute squares of its elements. The first term on the right side of the (5) is the fidelity term, and the second term is a regularization imposed on $\mathcal{X}$. The nonnegative parameter λ trades off the data fidelity and regularization. It is particularly worth noting that $\phi(\mathcal{X})$ is a regularizer expressing prior information tailored to spatially correlated eigenimages. An explicit definition of $\phi(\mathcal{X})$ is not given here because we design a CNN to learn the nonlocal self-similarity prior of eigenimages. Meanwhile, a proof of the equivalence of (4) and (5) can be found in reference [57].

Therefore, the optimization function (5) can be reformulated as the following formula:

$$\widehat{\mathcal{X}} = \psi_{\lambda\phi}(\mathcal{Y} \times_3 \mathbf{C}^T) \tag{6}$$

where $\psi_{\lambda\phi}(\mathcal{W}) = \arg\min(1/2)\|\mathcal{W} - \mathcal{X}\|_F^2 + \lambda\phi(\mathcal{X})$ is the so-called Moreau proximity operator of ϕ or denoising operator.

As images of natural scenes, an HSI is self-similar and similar patches can be found in nonlocal areas of HSI. The high correlation in grouped similar patches can be exploited to alleviate noise. We remark that as a linear combination of original bands, eigenimages contain the same spatial structure and are also self-similar (seen in Fig. 1). The high spatial correlation of eigenimages enables us to exploit the self-similarity of $\mathcal{X}$ as a type of regularization.

To exploit the self-similarity property of the eigenimages, we first generate a series of similar 3-D groups based on the feature images. What is more, the learning-based methods are used for denoising them in the next step. The detailed information can be found in Section III.

III. Proposed Method—NL-3DCNN

In this section, we detailed the proposed method. Benefiting from the great success of deep-learning-based methods in the field of image denoising recently, we designed a denoising method based on subspace representation and a 3-D CNN. Unlike the previous methods, which denoise the original hyperspectral image directly, we propose a new hyperspectral image denoising method based both on **N**on-**L**ocal self-similarity and 3-D **C**onvolutional **N**eural **N**etwork, termed **NL-3DCNN**, which can explore the spectral correlation of eigenimages and effectively reduce the dimension of features. The overall flowchart of the proposed method is shown in Fig. 1. The proposed method contains four parts which are listed as follows.

1) subspace identification of HSI;
2) 3-D similar patches' generation;
3) 3DCNN denoising of grouped patches;
4) recovery of the original HSI from estimated eigenimages;

A. Subspace Identification of HSI

The subspace identification step can be solved using singular value composition (SVD) of the spectral correlation

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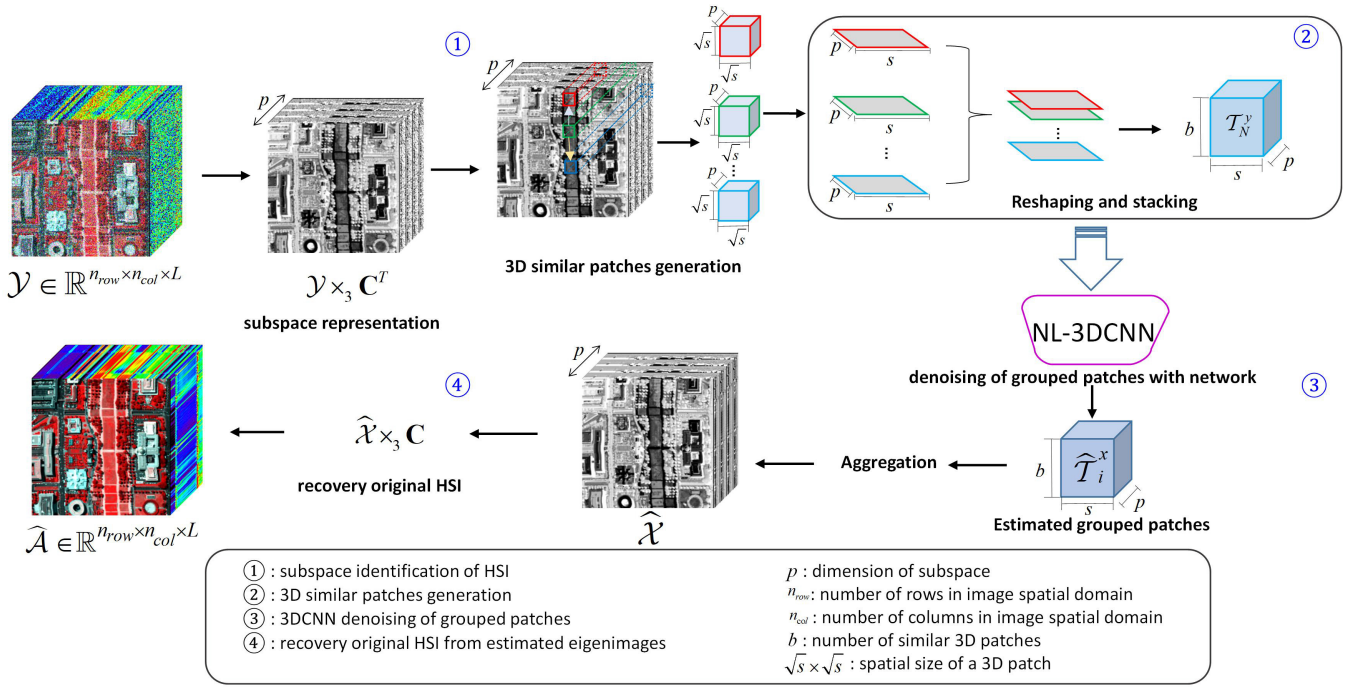


Fig. 1. Overall flowchart of the proposed NL-3DCNN.

matrix $\mathcal{Y}_{(3)}\mathcal{Y}_{(3)}^T/n$

$$[\mathbf{U}, \mathbf{S}, \mathbf{Z}] = \text{SVD}\left(\mathcal{Y}_{(3)}\mathcal{Y}_{(3)}^T/n\right) \tag{7}$$

$$\mathbf{C} = \mathbf{U}(:, \mathbf{1} : p) \tag{8}$$

where $\mathbf{U}$ and $\mathbf{Z}$ are the orthogonal matrices. Meanwhile, $\mathbf{S}$ is the singular value matrix, and $\mathbf{C}$ can be obtained from $\mathbf{U}$ with the first p columns.

B. 3-D Similar Patches Generation

After obtaining the orthogonal basis of $\mathbf{C}$, the eigenimages of the observed HSI can be obtained by computing $\mathcal{Y} \times_3 \mathbf{C}^T$. Let $\mathcal{X}_i' \in \mathbb{R}^{p \times \sqrt{s} \times \sqrt{s}}$ denote the 3-D patches extracted from the eigenimages, where $i \in \{1, \ldots, n\}$ is the index of the top-left pixel, and p and $\sqrt{s}$ are the dimension of subspace and the spatial width of the block, respectively. Correspondingly, the observation model of the ith 3-D block can be expressed as follows:

$$\mathcal{X}_i' = \mathcal{X}_i + \mathcal{N}_i' \tag{9}$$

where $\mathcal{X}_i \in \mathbb{R}^{p \times \sqrt{s} \times \sqrt{s}}$ is the clean block, and $\mathcal{N}_i' \in \mathbb{R}^{p \times \sqrt{s} \times \sqrt{s}}$ is the corresponding Gaussian noise.

To further explore the nonlocal self-similarity in eigenimages, we first get a 3-D patch with the dimension of $\mathcal{X}_i' \in \mathbb{R}^{p \times \sqrt{s} \times \sqrt{s}}$, and next sliding steps are used to process every next reference patch. p and $\sqrt{s}$ are the dimension of subspace and spatial size, respectively. Totally, $(b-1)$ similar patches are found according to the Frobenius distance, where b is the number of similar patches. Finally, these similar patches were reorganized into a third-order tenor $\mathcal{T}_i \in \mathbb{R}^{s \times b \times p}$, which can be redefined as follows:

$$\mathcal{T}_i^y = \mathcal{T}_i^x + \mathcal{T}_i^n \tag{10}$$

where $\mathcal{T}_i^y$, $\mathcal{T}_i^x$, and $\mathcal{T}_i^n$ are the tensors of similar patches corresponding to the observed noisy eigenimages, clean eigenimages, and noise, respectively. Also, the flowchart in Fig. 1 shows a generation process of the grouped similar patch.

C. 3DCNN Denoising of Grouped Patches

We propose a novel 3DCNN to alleviate noise in similar patches, $\mathcal{T}_i^y$, which is faster and more efficient than denoising on the original HSI.

In the aspect of network design, we modify the denoising convolutional neural networks (DnCNNs) [58] to make it suitable for HSI denoising. The proposed network framework is shown in Fig. 2. Compared with learning a mapping from noisy image to clean image, we consider residual learning to learn noise mapping from noisy patches to predict residual patches. Meanwhile, 3DCNN is used to explore the spatial–spectral information of similar patches. The details of the network structure are described below.

1) 3DCNN Model for Eigenimages Denoising: In Fig. 2, we show the proposed 3DCNN network architecture for eigenimage denoising. The network structure includes three parts: convolution layer, nonlinear activation layer, and residual output. We denote $\mathcal{T}_i^y \in \mathbb{R}^{s \times b \times p}$ and $\mathcal{T}_i^n \in \mathbb{R}^{s \times b \times p}$ as the input noisy patches and ground truth of noise, respectively. $\widehat{\mathcal{T}}_i^n \in \mathbb{R}^{s \times b \times p}$ is the residual image of the corresponding output. Our goal is to use the idea of residual learning to learn the distribution of noise from noisy similar patches $\mathcal{T}_i^y$, and then get clean similar patches $\widehat{\mathcal{T}}_i^x$. In particular, as 3-D convolutional network shows great potential for extracting deep representative features from high dimensions; here, we consider that the data of similar patch are a third-order tensor, so we use a 3-D convolution kernel with the size

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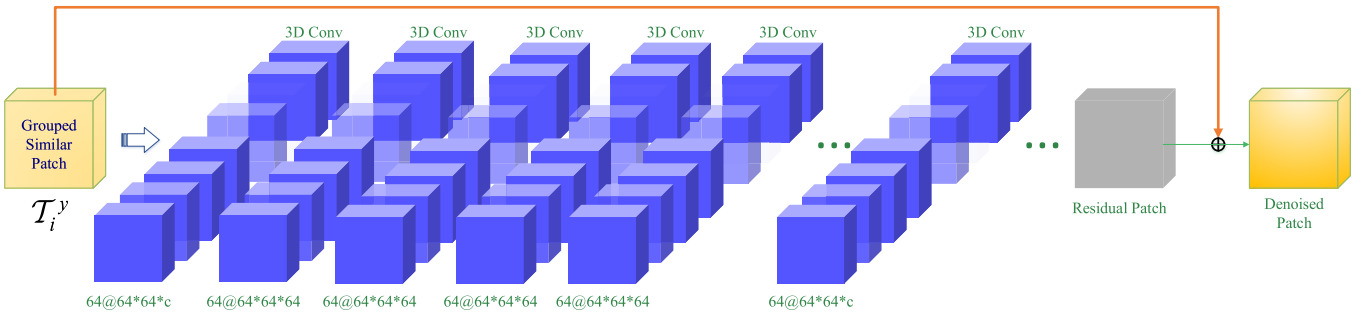


Fig. 2. Flowchart of the proposed 3-D CNN network architecture for eigenimage.

TABLE I
PROPOSED 3DCNN NETWORK ARCHITECTURE

Layer ID	Kernel Size	Kernels	Stride	Act-Function
3DConv(1)	$3 \times 3 \times 3$	64	(1,1,1)	ReLU
3DConv(2)	$3 \times 3 \times 3$	64	(1,1,1)	ReLU
3DConv(3)	$3 \times 3 \times 3$	64	(1,1,1)	ReLU
3DConv(4)	$3 \times 3 \times 3$	64	(1,1,1)	ReLU
3DConv(5)	$3 \times 3 \times 3$	64	(1,1,1)	ReLU
3DConv(6)	$3 \times 3 \times 3$	64	(1,1,1)	ReLU
3DConv(7)	$3 \times 3 \times 3$	64	(1,1,1)	ReLU
3DConv(8)	$3 \times 3 \times 3$	64	(1,1,1)	-

of $k \times k \times k$ to construct a 3-D CNN, aiming to explore the spectral and spatial information simultaneously. Thus, in the proposed network, K filters with the size of $k \times k \times k$ are used to learn the features of eigenimages $\mathcal{T}_i^y$, sliding across the width, height, and dimension with a stride $t \times t \times t$. Consequently, we can further have

$$F_R\left(\mathcal{T}_i^y\right) = H_{3\mathrm{DConv}_R}\left(\mathcal{T}_i^y\right) \tag{11}$$

where $H_{3\mathrm{DConv}_R}(.)$ denotes the Rth 3-D convolution operator. To enhance the expression ability of the network, each 3-D Conv layer is followed by a nonlinear activation layer. Hence, this layer performs a function $\mathcal{H}(.)$ to obtain the nonlinear relationships between the data $\mathcal{T}_i^y$

$$D_{\mathrm{Net}}\left(\mathcal{T}_i^y\right)^{(R)} = \mathcal{H}\left(F_{R-1}\left(\mathcal{T}_i^y\right)\right) \tag{12}$$

where $\mathcal{H}(.)$ is the activation layer, and many activation operators can be plugged into here. In this article, the rectified linear unit (ReLU) is considered as the activation function due to its computational efficiency. $D_{\mathrm{Net}}(.)$ is the proposed 3DCNN denoising network, which contains R layers (3DConv + ReLU), and the parameters of the proposed network structure are shown in Table I. Furthermore, the denoised image can be obtained by the difference between the noise image and the estimated noise

$$\widehat{\mathcal{T}}_i^x = \mathcal{T}_i^y - D_{\mathrm{Net}}\left(\mathcal{T}_i^y\right). \tag{13}$$

2) Loss Function: To measure the denoising performance of the eigenimage, several cost functions have been investigated in the remote sensing field, such as l_1 loss and l_2 loss. Because l_2 loss can efficiently penalize minor errors and maintain better convergence for Gaussian white noise, we adopt l_2 loss to measure the reconstruction accuracy of the proposed network. Specifically, considering the idea of residual learning, the loss function can be expressed as follows:

$$\zeta_1(\Theta) = \frac{1}{n_p}\sum_{i=1}^{n_p}\left\|D_{\mathrm{Net}}\left(\mathcal{T}_i^y, \Theta\right) - \mathcal{T}_i^n\right\|_2 \tag{14}$$

where $\|.\|_2$ is a standard method to compute the length of a vector in Euclidean space. $D_{\mathrm{Net}}(\mathcal{T}_i^y, \Theta)$ and n_p denote the estimated noise with the proposed network and the number of grouped similar pairs in one training batch, respectively. Θ is the parameter in the network, such as learning rate and batch size. $\mathcal{T}_i^n$ is the actual noise in the eigenimages $\mathcal{T}_i^y$.

D. Recovery of Original HSI From Estimated Eigenimages

For the part of recovery of the original HSI from estimated eigenimages, the denoised grouped 3-D similar patches using the proposed 3DCNN network contained in the tensor $\widehat{\mathcal{T}}_i^x$ are added back to their original location in the eigenimages. Meanwhile, the subspace basis, $\mathbf{C}$, is obtained from SVD. We can multiply $\widehat{\mathcal{X}}$ and $\mathbf{C}$ together by mode-3 product, and then we obtain the denoised hyperspectral image

$$\widehat{\mathcal{A}} = \widehat{\mathcal{X}} \times_3 \mathbf{C}. \tag{15}$$

IV. EXPERIMENTS ON SYNTHETIC DATA

In this section, a series of simulated datasets based on Washington dc MAll were generated to verify the effectiveness of the proposed NL-3DCNN method. In this article, Gaussian white noise with different intensities was added.

A. Experimental Setting

1) Implementation Details: As mentioned in Section III, there are mainly four steps in the proposed NL-3DCNN method. The original HSI is represented as an eigenimage first, and then a series of nonlocal similar patches are searched by Frobenius distance in the eigenimage and reshaped to a 3-D tensor subsequently. Next, we propose a 3DCNN network for estimating the spatial–spectral information of similar patches. By finding the position of similar patches in the feature image, the denoised image can be obtained.

More specifically, we implement the NL-3DCNN network using the Pytorch framework, which is trained on two TITAN

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TABLE II
DENOISING RESULTS OF THE PROPOSED METHOD AND STATE-OF-THE-ART METHODS ON WASHINGTON DC MALL

NoiseLevel		Metric	Noisy Image	LRTA	BM4D	LRMR	HSID-CNN	HSI-SDeCNN	QRNN3D	NG-Meet	NL-3DCNN
				[60]	[14]	[26]	[52]	[53]	[46]	[18]	(Proposed)
Case1	0.02	MPSNR (dB)	34.1533	38.9607	41.1030	42.6349	41.6857	39.9717	41.7746	41.9557	**42.8831**
		MSSIM	0.9819	0.9925	0.9962	**0.9972**	0.9966	0.9954	0.9967	0.9967	0.9968
		MFSIM	0.9817	0.9919	0.9959	**0.9966**	0.9959	0.9950	0.9960	0.9962	0.9965
		SAM	0.0741	0.0474	0.0342	**0.0295**	0.0320	0.0391	0.0320	0.0315	0.0312
		Time (s)	0	1	148	95	3	1	8	41	211
	0.1	MPSNR (dB)	20.1739	30.8470	31.0498	33.3851	33.0401	33.3501	34.3060	35.3967	**35.8679**
		MSSIM	0.7604	0.9643	0.9684	0.9813	0.9799	0.9817	0.9842	0.9885	**0.9892**
		MFSIM	0.8587	0.9662	0.9718	0.9804	0.9805	0.9832	0.9830	0.9869	**0.9878**
		SAM	0.3069	0.0990	0.0910	0.0777	0.0745	0.0692	0.0637	0.0562	**0.0529**
		Time (s)	0	2	147	94	3	1	7	84	130
	0.2	MPSNR (dB)	14.1533	26.9101	26.6829	28.8287	28.9766	29.4871	30.8822	31.8599	**32.1988**
		MSSIM	0.5065	0.9261	0.9204	0.9529	0.9538	0.9595	0.9683	0.9767	**0.9782**
		MFSIM	0.7465	0.9358	0.9397	0.9600	0.9608	0.9666	0.9693	0.9757	**0.9775**
		SAM	0.5182	0.1283	0.1294	0.1118	0.1080	0.0958	0.0842	0.0755	**0.0708**
		Time (s)	0	2	148	95	3	1	7	84	91
	0.3	MPSNR (dB)	10.6315	24.6643	24.1912	26.2260	26.7706	27.1978	27.9317	29.2955	**29.6928**
		MSSIM	0.3419	0.8866	0.8660	0.9234	0.9274	0.9346	0.9408	0.9614	**0.9633**
		MFSIM	0.6623	0.9103	0.9085	0.9415	0.9445	0.9501	0.9516	0.9616	**0.9657**
		SAM	0.6754	0.1452	0.1565	0.1342	0.1311	0.1160	0.1171	0.0927	**0.0890**
		Time (s)	0	2	148	95	3	1	8	83	36
	0.4	MPSNR (dB)	8.1327	23.0389	22.5104	24.3045	25.3044	25.5812	26.0490	27.8551	**28.6417**
		MSSIM	0.2385	0.8449	0.8103	0.8902	0.9013	0.9083	0.9155	0.9476	**0.9559**
		MFSIM	0.5960	0.8873	0.8785	0.9235	0.9294	0.9339	0.9386	0.9511	**0.9604**
		SAM	0.7959	0.1606	0.1777	0.1550	0.1460	0.1328	0.1411	0.1040	**0.0911**
		Time (s)	0	2	148	94	3	1	8	84	32

RTX graphics processing units (GPUs). The grouped 3-D similar patches were fed to the network with a batch size of 16. Adam optimizer [59] is used to optimize the network. The number of feature maps was set as 64, which is the same in [58]. The number of training epochs and the initial learning rate were set to 35 and 1×10^{-4}, respectively.

2) Simulation of Datasets: The Washington dc Mall, which was collected by Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor, was used as a clean image in simulated experiments. The image contains 1280×303 pixels with 191 spectral bands, ranging from 0.4 to 2.5 μm over the visible and infrared spectrum. Before the experiment, pixel values of the image were normalized to the range of [0, 1]. The noisy HSIs for simulated experiments are generated. In general, different cases of noise were directly added into the original hyperspectral clean image to generate noisy HSIs. Rather than feed the original image into the network, we feed the grouped similar patches into the network, and the patch matching step is isolated from the training process. More specifically, the noisy HSIs are projected on the subspace first. Then, the grouped similar noisy patches are generated from noisy eigenimages (shown in Fig. 1). Likewise, the grouped similar clean patches are generated from clean eigenimages. As for the testing samples, the generation is similar to the training samples; we can obtain the grouped similar patches according to the image of noisy eigenimages, and then we feed these grouped similar patches into the network to obtain the denoised grouped similar patches. After that, we can obtain the denoised eigenimages, and according to (15), we can get the denoised HSIs. Two kinds of additive noise are considered in the simulated experiments as follows:

Case 1 (Gaussian i.i.d. Noise): $\mathbf{n}_i \sim \mathcal{N}(0, \sigma^2\mathbf{I})$ with $\sigma \in \{0.02, 0.1, 0.2, 0.3, 0.4\}$

Case 2 (Gaussian Non-i.i.d. Noise): $\mathbf{n}_i \sim \mathcal{N}(0, \mathbf{D^2})$ where $\mathbf{D}$ is a diagonal matrix with diagonal elements sampled from uniform distributions, i.e., U (0.02, 0.04), U (0.1, 0.2), and U (0.2, 0.4).

A subregion of Washington dc Mall data with the size of $1080 \times 303 \times 191$ is used to train the 3DCNN network. The rest part of the Washington dc Mall, with the size of $200 \times 200 \times 191$, and the subimage of Pavia Center over Pavia, northern Italy, with the size of $200 \times 200 \times 102$, are used to test the proposed method, respectively.

B. Comparison and Evaluation

First, we verify the effectiveness of the proposed NL-3DCNN method in the simulated test data and compare it with several state-of-the-art HSI denoising methods, including lower rank tensor approximation (LRTA) [60], block matching with 4-D filtering (BM4D) [14], low-rank matrix recovery (LRMR) [26], HSID-CNN [52], HSI-SDeCNN [53], QRNN3D [46], and non-local meets global (NG-Meet) [18]. Meanwhile, to quantitatively evaluate all the methods, this article adopts four commonly used image reconstruction evaluation indexes, namely, peak signal-to-noise ratio (PSNR), structural similarity (SSIM) [61], spectral angular mapper (SAM), and feature similarity (FSIM) [62]. All the indicators (PSNR, SSIM, SAM, and FSIM) are calculated band by band. The mean PSNR (MPSNR), mean SSIM (MSSIM), and mean FSIM (MFSIM) of the proposed method and all the comparison methods are recorded in Tables II and III.

C. Parameter Setting

In this experiment, we analyzed the parameters used in all the algorithms. For BM4D, the parameter of the standard deviation of Gaussian noise was set to the true value.

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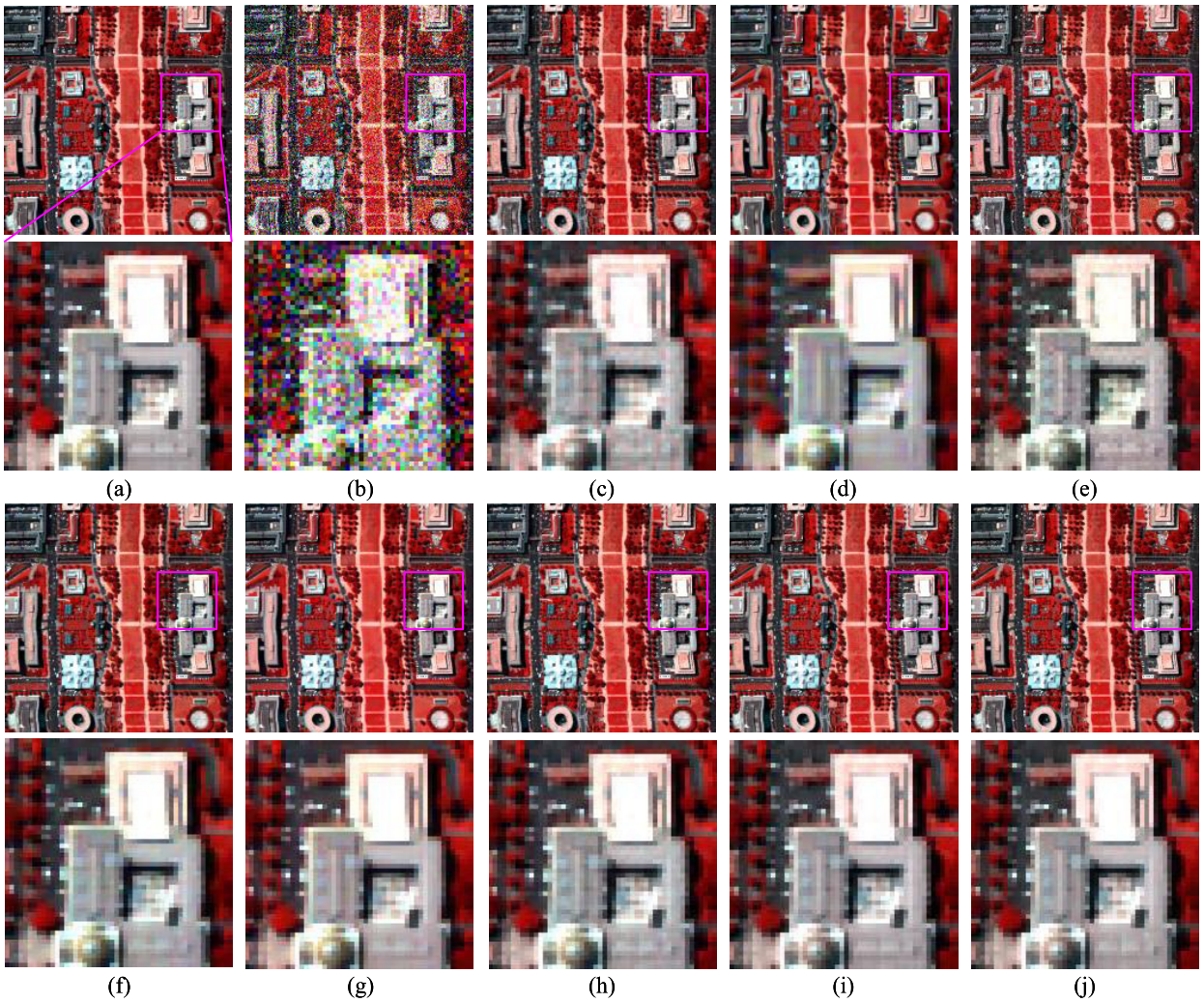


Fig. 3. False color denoising results (bands 57, 27, and 17) with $\sigma_n = 0.2$. (a) Clean image. (b) Noisy image. (c) LRTA. (d) BM4D. (e) LRMR. (f) HSID-CNN. (g) HSIDeCNN. (h) QRNN3D. (i) NG-Meet. (j) Proposed NL-3DCNN method.

TABLE III

DENOISING RESULTS OF THE PROPOSED METHOD AND STATE-OF-THE-ART METHODS ON WASHINGTON DC MALL (CASE 2)

NoiseLevel		Metric	Noisy Image	LRTA	BM4D	LRMR	HSID-CNN	HSI-SDeCNN	QRNN3D	NG-Meet	NL-3DCNN
				[60]	[14]	[26]	[52]	[53]	[46]	[18]	(Proposed)
Case2	(0.02-0.04)	MPSNR (dB)	30.7963	36.5171	36.1052	40.0780	40.7707	38.4310	39.7782	40.2496	**41.6093**
		MSSIM	0.9613	0.9882	0.9905	0.9949	**0.9962**	0.9938	0.9949	0.9951	0.9960
		MFSIM	0.9657	0.9879	0.9891	0.9941	**0.9961**	0.9933	0.9939	0.9944	0.9956
		SAM	0.1100	0.0598	0.0575	0.0407	**0.0331**	0.0447	0.039	0.0388	0.0342
		Time (s)	0	1	148	95	3	1	7	41	211
	(0.1-0.2)	MPSNR (dB)	16.8174	28.8781	26.4649	30.2146	30.8861	30.3333	32.3188	33.4008	**33.7976**
		MSSIM	0.6255	0.9488	0.9277	0.9644	0.9701	0.9649	0.9763	0.9826	**0.9835**
		MFSIM	0.7996	0.9534	0.9416	0.9662	0.9744	0.9690	0.9764	0.9817	**0.9828**
		SAM	0.4266	0.1137	0.1365	0.1104	0.0859	0.0978	0.0742	**0.0667**	0.0680
		Time (s)	0	2	148	94	3	1	7	93	130
	(0.2-0.4)	MPSNR (dB)	10.7968	24.9472	22.4370	25.3241	26.5015	25.9497	28.4007	29.5535	**30.3870**
		MSSIM	0.3512	0.8926	0.8399	0.9072	0.9260	0.9134	0.9473	0.9623	**0.9688**
		MFSIM	0.6653	0.9131	0.8927	0.9281	0.9438	0.9358	0.9553	0.9638	**0.9704**
		SAM	0.6829	0.1448	0.1898	0.1813	0.1285	0.1421	0.1065	0.0912	**0.0798**
		Time (s)	0	2	147	94	3	1	7	84	91

The parameters in LRMR and NG-Meet were hand-tuned to get the optimal performance. Considering that HSID-CNN, HSI-SDeCNN, and QRNN3D are deep-learning-based methods, we keep the same training dataset with the proposed NL-3DCNN to verify the effectiveness of the algorithm. The dimension of subspace, p, is a key hyperparameter, which can be estimated by the Hysime method [63] or gained from experience. Empirically, the dimension of subspace should be set

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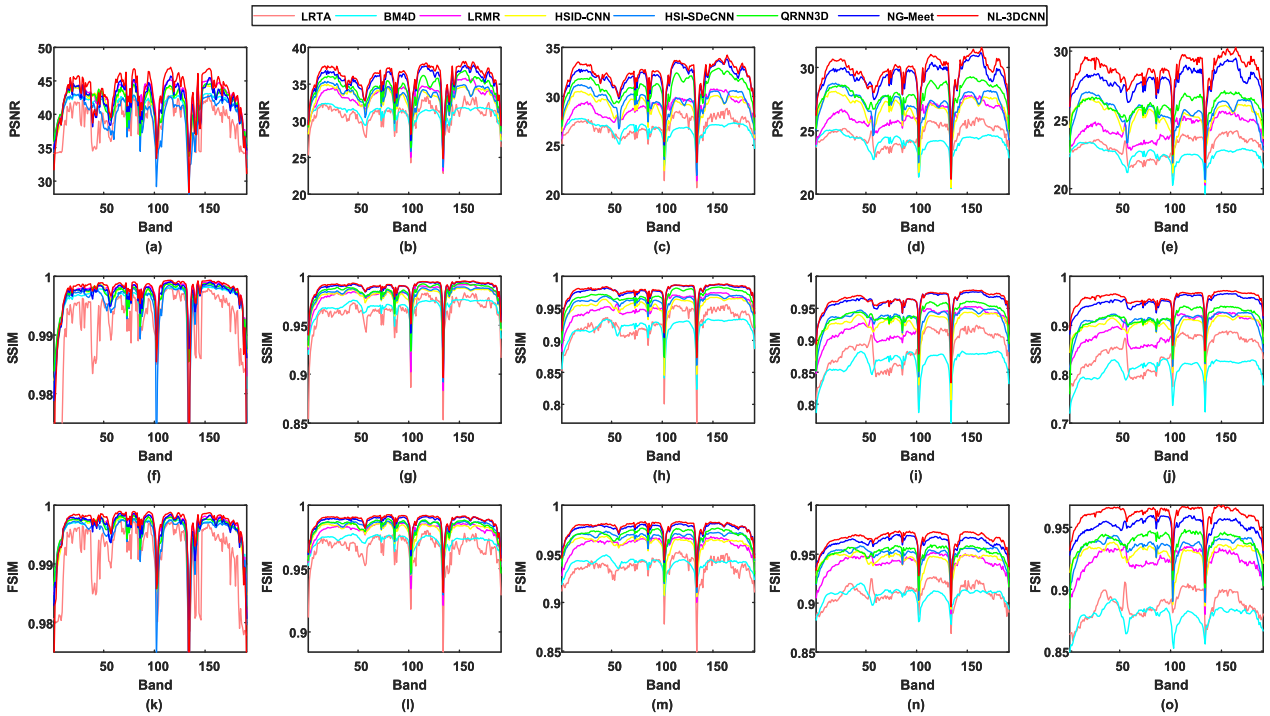


Fig. 4. (a), (f), and (k) Results of PSNR, SSIM, and FSIM, respectively ($\sigma = 0.02$). (b), (g), and (f) Results of PSNR, SSIM, and FSIM, respectively ($\sigma = 0.1$). (c), (h), and (m) Results of PSNR, SSIM, and FSIM, respectively ($\sigma = 0.2$). (d), (i), and (n) Results of PSNR, SSIM, and FSIM, respectively ($\sigma = 0.3$). (e), (j), and (o) Results of PSNR, SSIM, and FSIM, respectively ($\sigma = 0.4$).

as a large value (say 20) when the noise intensity is small. For images with severer noise, the dimension of subspace should be set as a small value (say 6). For the proposed NL-3DCNN method, the dimension of subspace, p, for different levels of Gaussian white noise ($\sigma_n = \{0.02, 0.1, 0.2, 0.3, 0.4\}$) was set to 20, 16, 10, 7, and 6, respectively. In all the experiments, the number of similar 3-D patches and the spatial size of a patch were set to 64 and 8×8, respectively. Therefore, for every noisy level, 39 492 grouped similar patches were obtained to train the 3DCNN network.

D. Denoising Results and Analysis

We first numerically illustrate the denoising performance of the proposed NL-3DCNN method and compare it with other state-of-the-art methods. The denoising results of Washington dc Mall in terms of four indicators (namely, MPSNR, MSSIM, MFSIM, and SAM) are displayed in Tables II and III. The best of these results are shown in bold. As shown in Tables II and III, in the case of five levels of Gaussian i.i.d. noise and three levels of Gaussian non-i.i.d. noise, all the methods, including LRTA, BM4D, LRMR, HSID-CNN, HSI-SDeCNN, QRNN3D, NG-Meet, and the proposed NL-3DCNN, can significantly improve the image quality. The proposed NL-3DCNN method obtains the best denoising performance in most cases.

To visually compare the denoising performance among different methods, we show the false color denoising results of bands 57, 27, and 17 in the case with $\sigma_n = 0.2$ in Fig. 3. More specifically, Fig. 3(a) and (b) shows the clean image and the noisy image, respectively. Fig. 3(c)–(j) displays the denoising results of different methods. To more intuitively see the differences between different methods, we clipped a small subarea from the original image, as shown in Fig. 3(a). Through analysis, we can see that the LRTA and LRMR cannot completely remove the noise in the image; there still exists residual noise. Meanwhile, BM4D gets oversmoothed results. However, BM4D is a nonlocal-based method, which is designed for high-dimensional images, such as video. Therefore, the BM4D method has some limitations when applied to HSI processing. HSID-CNN, HSI-SDeCNN, and QRNN3D are the deep-learning-based methods; all of them can obtain a better denoising results. Furthermore, NG-Meet and our proposed method perform better than HSID-CNN, HSI-SDeCNN, and QRNN3D. Both the NG-Meet and the proposed NL-3DCNN methods are low-rank-based subspace denoiser, and both of them focus on Gaussian noise removal in the images. Meanwhile, there are two significant differences between the NG-Meet method and the NL-3DCNN method. The first significant difference is that the subspace basis ($\mathbf{E}$) of NG-Meet is updated iteratively, while NL-3DCNN only performs one time for subspace extraction from noisy images. Furthermore, the nonlocal self-similarity is explored using the weighted nuclear norm technique for NG-Meet, while the proposed NL-3DCNN uses a network-based technique, termed 3DCNN. The experiments show that 3DCNN can perform better numerically and visually compared with NG-Meet to explore the nonlocal self-similarity in the images.

Fig. 4 illustrates the denoising results for each band of Washington dc MAll in terms of PSNR, SSIM, and FSIM. Fig. 4(a)–(e) shows the PSNR values over bands in images with noise level $\sigma_n = \{0.02, 0.1, 0.2, 0.3, 0.4\}$, respectively. The red lines represent denoising results of the proposed method. It is clear that our proposed performs better in most cases than other methods.

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TABLE IV
DENOISING RESULTS OF THE PROPOSED METHOD AND STATE-OF-THE-ART METHODS ON THE PAVIA CENTER DATASET

NoiseLevel		Metric	Noisy Image	LRTA	BM4D	LRMR	HSID-CNN	HSI-SDeCNN	QRNN3D	NG-Meet	NL-3DCNN
				[60]	[14]	[26]	[52]	[53]	[46]	[18]	(Proposed)
Case1	0.02	MPSNR (dB)	34.0315	41.0594	42.6987	44.4953	43.5166	43.7197	43.3796	44.9081	**45.3373**
		MSSIM	0.9125	0.9741	0.9775	0.9877	0.9851	0.9846	0.9858	**0.9894**	0.9873
		MFSIM	0.9622	0.9889	0.9896	0.9944	0.9933	0.9936	0.9936	**0.9954**	0.9946
		SAM	0.1610	0.0763	0.0594	0.0516	0.0579	0.0553	0.0578	**0.0467**	0.0501
		Time (s)	0	1	79	49	2	1	7	17	211
	0.1	MPSNR (dB)	20.0521	33.4692	34.0190	33.0486	35.4542	36.3919	34.8436	35.8590	**36.6327**
		MSSIM	0.3747	0.9062	0.9121	0.8987	0.9362	0.9469	0.9366	0.9394	**0.9496**
		MFSIM	0.7064	0.9466	0.9443	0.9537	0.9680	0.9723	0.9673	0.9706	**0.9737**
		SAM	0.5885	0.1266	0.1180	0.1693	0.1159	0.0979	0.1608	0.1111	**0.0944**
		Time (s)	0	2	79	50	2	1	7	28	130
	0.2	MPSNR (dB)	14.0315	29.7735	30.2896	28.6564	31.8571	32.8419	31.3157	32.5874	**33.1191**
		MSSIM	0.1471	0.8203	0.8257	0.8013	0.8760	0.8972	0.8764	0.8934	**0.9059**
		MFSIM	0.5225	0.8916	0.8846	0.8995	0.9331	0.9407	0.9327	0.9431	**0.9530**
		SAM	0.8830	0.1459	0.1622	0.2458	0.1550	0.1295	0.1739	0.1375	**0.1305**
		Time (s)	0	1	79	59	2	1	8	28	91
	0.3	MPSNR (dB)	10.5097	27.5618	28.2389	26.7285	29.9195	30.7236	27.9055	30.7783	**30.8190**
		MSSIM	0.0721	0.7327	0.7484	0.7172	0.8216	0.8462	0.7655	0.8525	**0.8602**
		MFSIM	0.4169	0.8471	0.8371	0.8671	0.9022	0.9077	0.8895	0.9186	**0.9284**
		SAM	1.0539	0.1592	0.1900	0.2763	0.1782	0.1548	0.2586	0.1522	**0.1509**
		Time (s)	0	1	79	57	2	1	7	28	36
	0.4	MPSNR (dB)	8.0109	26.1036	26.8675	25.1175	28.8897	29.2280	26.7743	29.4924	**29.5717**
		MSSIM	0.0406	0.6564	0.6815	0.6344	0.7859	0.7975	0.7183	0.8143	**0.8214**
		MFSIM	0.3478	0.8124	0.7999	0.8349	0.8775	0.8772	0.8674	0.8960	**0.9085**
		SAM	1.1613	0.1685	0.2091	0.3080	0.1852	0.1752	0.2746	0.1627	**0.1578**
		Time (s)	0	1	78	58	2	1	7	28	32

TABLE V
ABLATION STUDY BY COMBINING THE NONLOCAL SELF-SIMILARITY AND 3DCNN

NoiseLevel		Parameters	Metric	Noisy	"3DCNN+WholeImage"	"3DCNN+Eigenimages"	"3DCNN+Nonlocal Self-similarity"
Case2	(0.02-0.04)	0.67M	MPSNR (dB)	30.7963	40.4521	40.7331	**41.6093**
			MSSIM	0.9613	0.9957	0.9954	**0.9960**
			MFSIM	0.9657	0.9952	0.9945	**0.9956**
			SAM	0.1100	0.0346	0.0375	**0.0342**

We report the running time of all the methods in Table II. It is clear that deep-learning-based methods (HSID-CNN, HSI-SDeCNN, and QRNN3D) require less inference time in most cases compared with the model-based methods (LRMR and NG-Meet). Meanwhile, it can also be seen that the nonlocal-based methods (BM4D, NG-Meet, and NL-3DCNN) always require more running time than other methods. For the proposed NL-3DCNN method, the running time reduces gradually as the dimension of the subspace decreases. The main reason is that the less dimension of the subspace will greatly reduce the size of images required for the network. (For example, once the network is trained, the inference time of the data $H \times W \times p_1$ is longer than the data $H \times W \times p_2$ ($p_1 > p_2$), where H and W are the height and width of the image, respectively.

In addition, we trained the NL-3DCNN network in Washington dc and directly test the denoising results on the Pavia center dataset to verify the robustness of the proposed NL-3DCNN and compare it with other methods. As shown in Table IV, HSI-SDeCNN is more robust than HSID-CNN and QRNN3D. Meanwhile, NL-3DCNN performs better than other purely deep-learning-based methods.

E. Ablation Study

To the verify the effectiveness of 3-D CNN and nonlocal self-similarity, a series of ablation studies were produced (shown in Table V). The ablation study is to show the effectiveness of combining the 3DCNN and nonlocal self-similarity. More specifically, Table V shows the denoising results of case 2 with the metric of MPSNR, MSSIM, MFSIM, and SAM. If we directly feed the original image into 3DCNN ("3DCNN + WholeImage"), the reconstructed image has an MPSNR of 40.4521 dB. If we feed the eigenimages into the network ("3DCNN + Eigenimages"), the corresponding MPSNR is 40.7331 dB. Compared with feeding the original image into 3DCNN, the denoising results are significantly improved (41.6093 dB) by feeding nonlocal similar patches into the network ("3DCNN + Nonlocal Self-similarity").

To gain more insight into the denoising process of the proposed network NL-3DCNN, we visualized the features extracted by the middle layers. Fig. 5 depicts the feature maps in Layer1, Layer3, Layer5, and Layer7 of the NL-3DCNN network when a patch of observed HSI is fed into the network. It is clear that with the increase in the number of network layers, the noise of the feature layer is obviously reduced. Fig. 6 shows the average epoch loss of the training process with $\sigma_n = 0.2$, and the proposed network converges after 35 training epochs.

To sum up, the proposed NL-3DCNN performs better than the state-of-the-art methods. Compared with HSID-CNN denoising on the original HSIs, this article proposes a denoising network based on grouped similar patches, which can better explore the nonlocal self-similarity of the eigenimage

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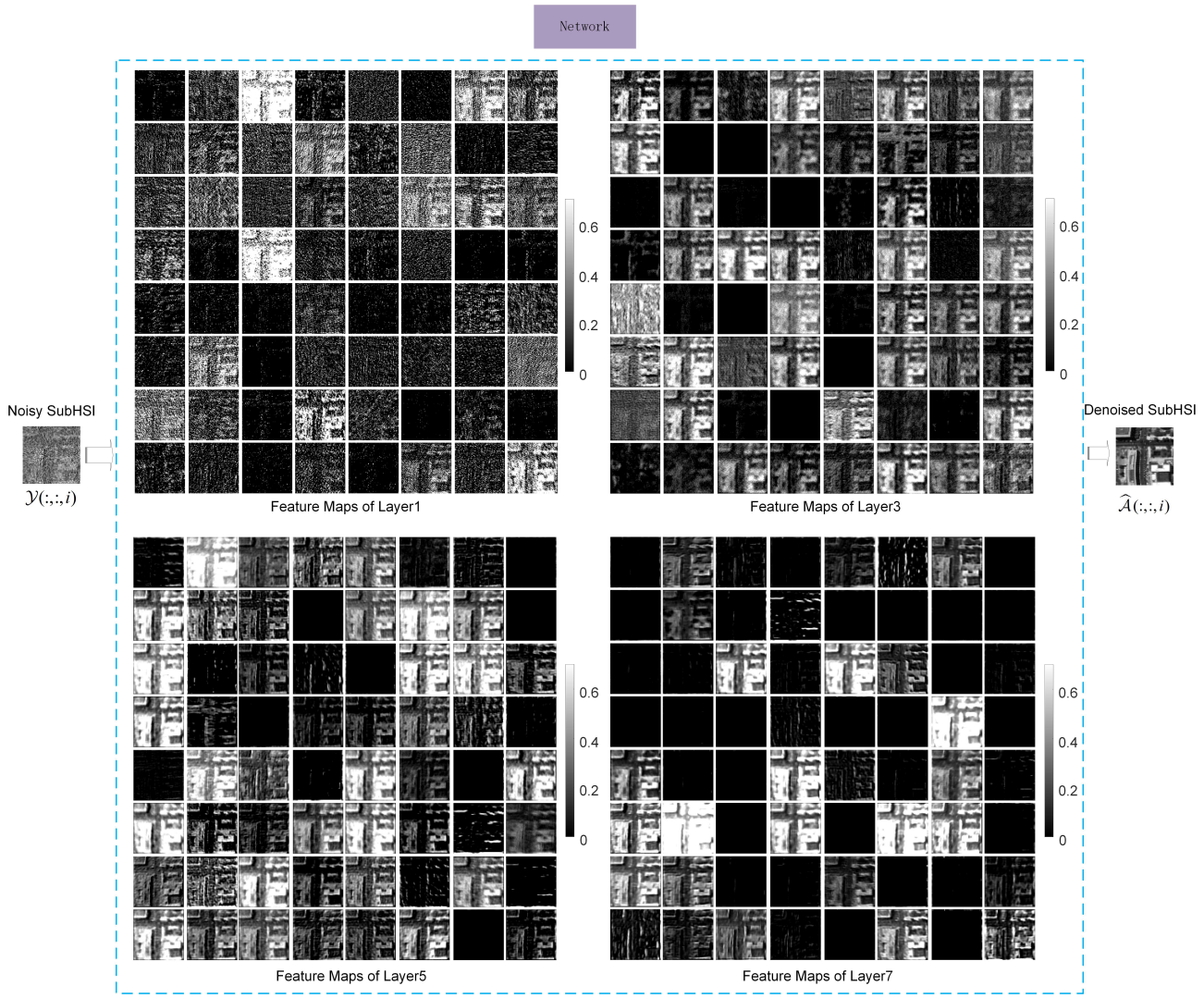


Fig. 5. Feature maps in Layer1, Layer3, Layer5, and Layer7 of the NL-3DCNN network when a patch of observed HSI is fed into the network.

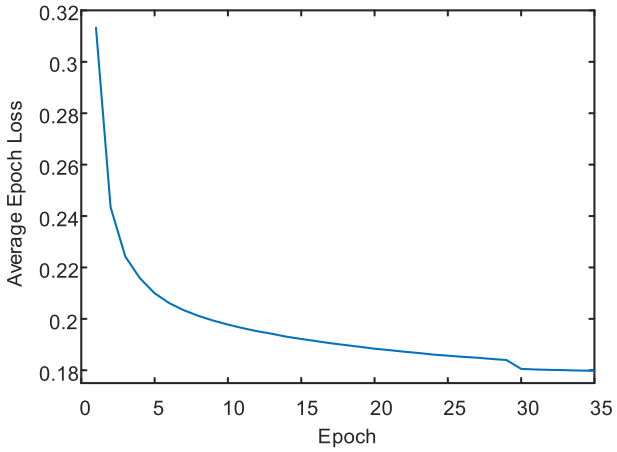


Fig. 6. Average epoch training loss with $\sigma_n = 0.4$.

and keep a balance between filtering and detail preservation of HSIs.

V. Experiments on Real Dataset

In this section, to verify the proposed algorithm's effectiveness, we use two real urban HSIs, the Urban dataset and ZY1E dataset, to conduct experiments. In real experiments, the way to train 3DCNN is similar to HSID-CNN [52]. The difference is that the standard deviations of Gaussian noise in the two real HSIs were estimated using the HySime method [63], and then the estimated noise was added to clean dc image to generate paired noisy clean training samples to train 3DCNN. The first real dataset was whitened before denoising [57].

A. Urban Real Dataset

The first real dataset used in this article is Urban, which is widely used in the field of HSI denoising. The Urban dataset, obtained by HYDICE sensor over an urban area, has 210 spectral bands with the size of 307 × 307 [shown in Fig. 7(a)].

The parameters in all the comparison methods were manually set to optimal values. For the proposed NL-3DCNN, the dimension of subspace, p, was set to 6. This article visualizes three bands of Urban that are seriously polluted by noise for comparison, including band 104 and band 139. Specifically, the denoising results of Urban are shown in Figs. 8 and 9, respectively. Fig. 8(a) shows that there exists Gaussian noise and stripe noise in the horizontal and

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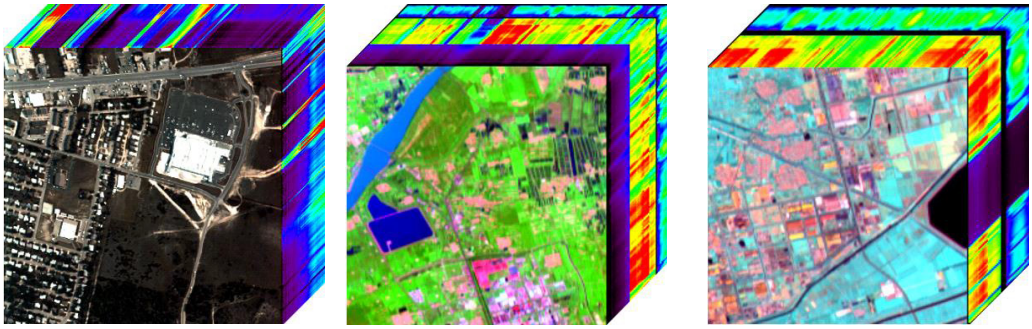


Fig. 7. Real HSI datasets used in the experiments. (a) Urban real dataset. (b) Subimage1 of the ZY1E real dataset. (c) Subimage2 of the ZY1E real dataset.

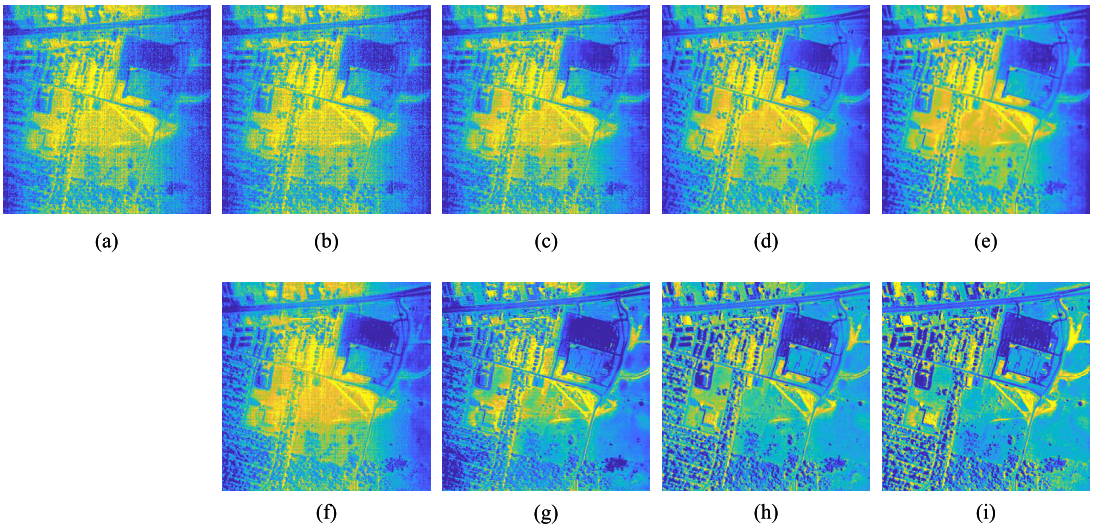


Fig. 8. (a) Original image with band 104. (b) LRTA. (c) BM4D. (d) LRMR. (e) HSID-CNN. (f) HSI-SDeCNN. (g) QRNN3D. (h) NG-Meet. (i) Proposed NL-3DCNN method.

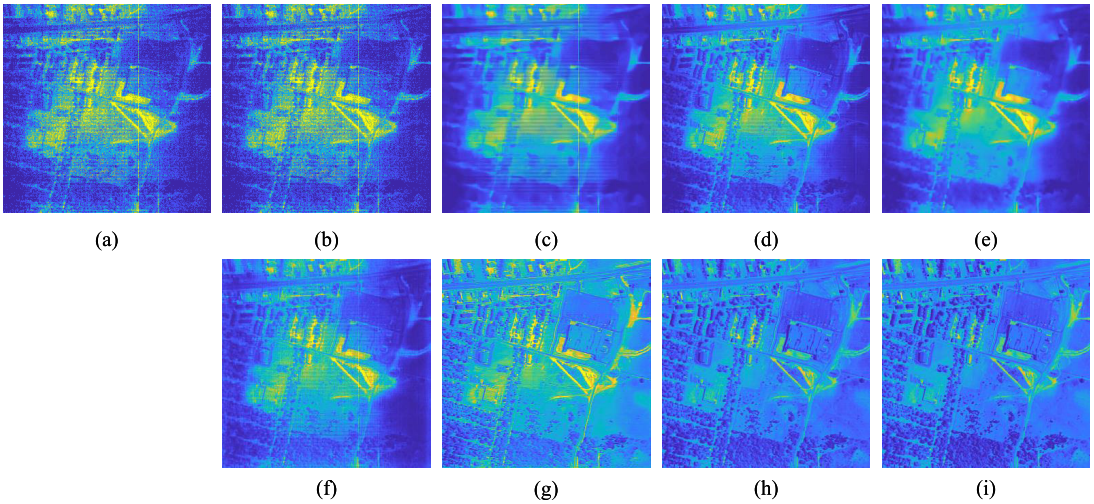


Fig. 9. (a) Original image with band 139. (b) LRTA. (c) BM4D. (d) LRMR. (e) HSID-CNN. (f) HSI-SDeCNN. (g) QRNN3D. (h) NG-Meet. (i) Proposed NL-3DCNN method.

vertical directions. It is clear that LRTA, BM4D, LRMR, HSID-CNN, and HSI-SDeCNN cannot remove the noise totally. QRNN3D, NG-Meet, and the proposed NL-3DCNN perform better visually. Band 139 has Gaussian noise, deadlines, and nonuniform illumination. As shown in Fig. 9, only QRNN3D, NG-Meet, and the proposed NL-3DCNN method

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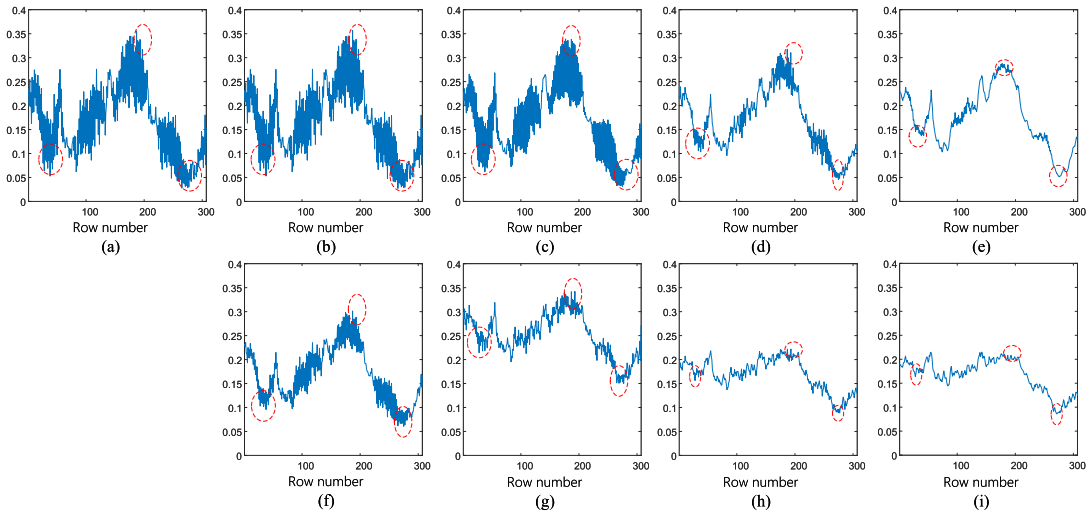


Fig. 10. Horizontal mean profiles of band 139 in the Urban dataset. (a) Noisy. (b) LRTA. (c) BM4D. (d) LRMR. (e) HSID-CNN. (f) HSI-SDeCNN. (g) QRNN3D. (h) NG-Meet. (i) Proposed NL-3DCNN method.

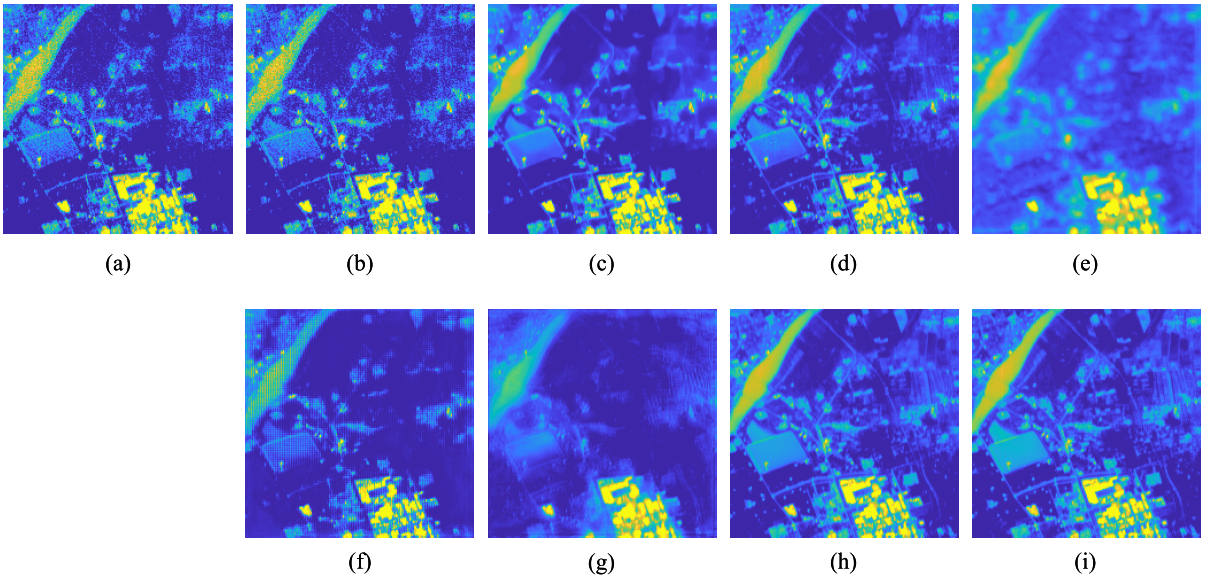


Fig. 11. (a) Original image with band 1 (Subimage1). (b) LRTA. (c) BM4D. (d) LRMR. (e) HSID-CNN. (f) HSI-SDeCNN. (g) QRNN3D. (h) NG-Meet. (i) Proposed NL-3DCNN method.

can remove deadlines and correct the spatial illumination well. Furthermore, the corresponding horizontal mean profiles of band 139 in the Urban dataset are shown in Fig. 10, where NL-3DCNN obtains a smoother horizontal mean profile than QRNN3D and NG-Meet, implying that less noise leaves in the result of NL-3DCNN.

B. ZY1E Real Dataset

The second real dataset used in this article is the ZY1E hyperspectral remote sensing image. The image has 166 spectral bands with a spatial resolution of 30 m. The image contains complex noises such as Gaussian noise and striping noise. After removing several bands with very low SNR, 150 bands are retained for testing. In this article, two subareas of size 200 × 200 [shown in Fig. 7(b) and (c)] are extracted to verify the effectiveness of the proposed NL-3DCNN method.

Similarly, the parameters in all the comparison methods were set to optimal values manually. For the proposed NL-3DCNN, the dimension of subspace, p, was set to 6. Meanwhile, three noisy bands (1, 114, and 128) are presented in first subarea to illustrate the effectiveness of the proposed method. The results in Figs. 11–13 show that the proposed NL-3DCNN and NG-Meet can effectively remove Gaussian noise and slight stripe noise in the image. More specifically, as shown in Fig. 13, zoomed areas of its denoising results show that our proposed NL-3DCNN method performs better than NG-Meet. LRTA, BM4D, and LRMR cannot remove the noise thoroughly. By comparing the deep-learning-based methods, namely, HSID-CNN, HSI-SDeCNN, QRNN3D, and NL-3DCNN, it is easy to see that HSID-CNN and QRNN3D get oversmoothed denoising results to deal with Gaussian noise, and the proposed NL-3DCNN can keep the detailed texture information of the image very well. In addition, the

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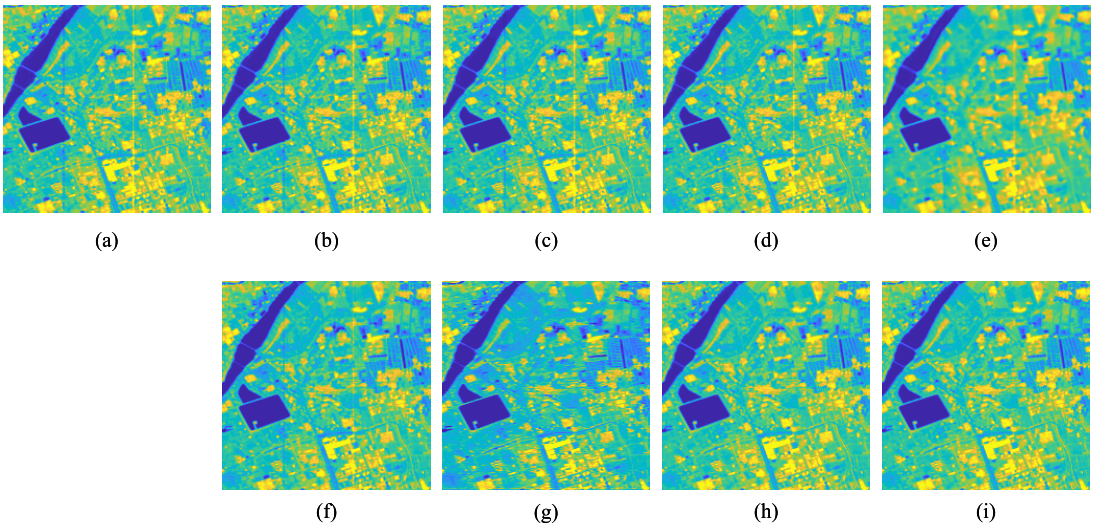


Fig. 12. (a) Original image with band 114 (Subimage1). (b) LRTA. (c) BM4D. (d) LRMR. (e) HSID-CNN. (f) HSI-SDeCNN. (g) QRNN3D. (h) NG-Meet. (i) Proposed NL-3DCNN method.

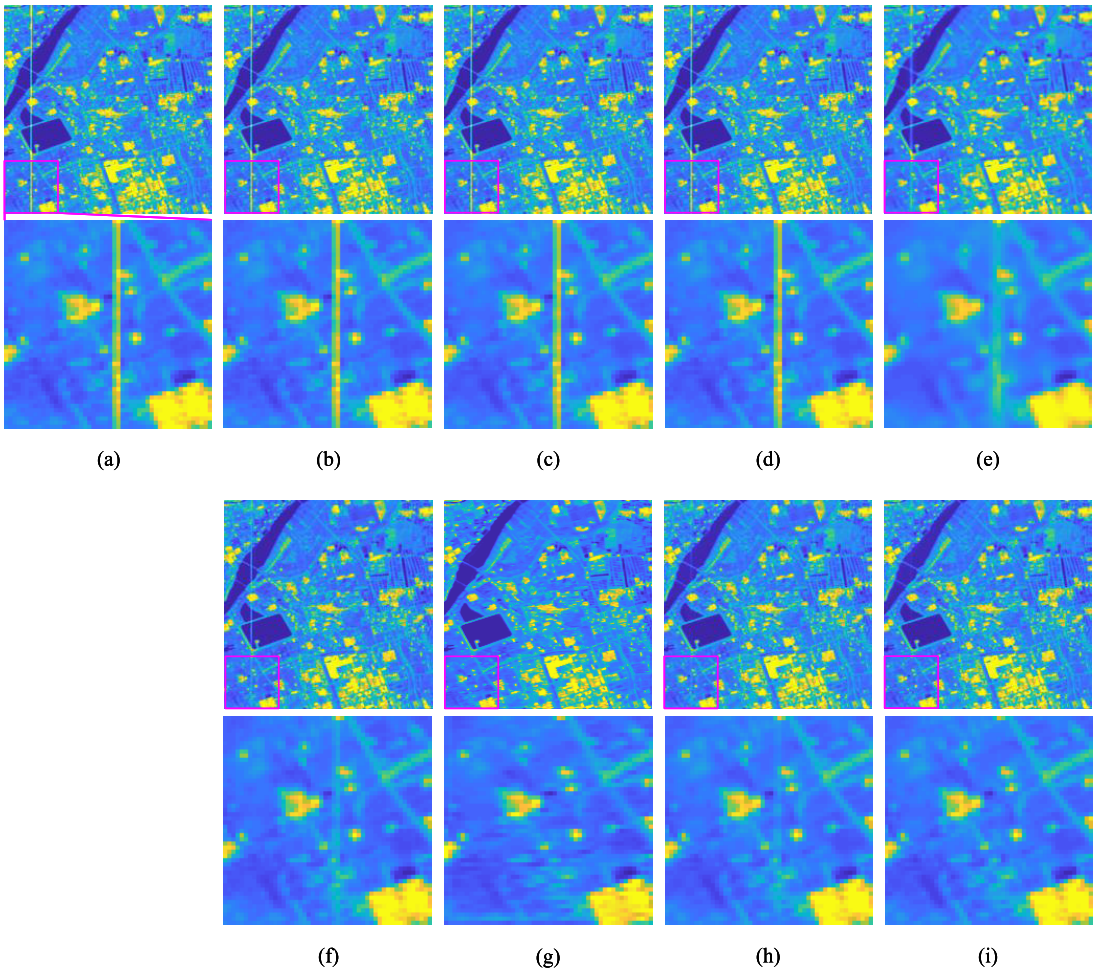


Fig. 13. (a) Original image with band 128 (Subimage1). (b) LRTA. (c) BM4D. (d) LRMR. (e) HSID-CNN. (f) HSI-SDeCNN. (g) QRNN3D. (h) NG-Meet. (i) Proposed NL-3DCNN method.

corresponding vertical mean profiles of band 128 in the ZY1E dataset are shown in Fig. 14, where NL-3DCNN obtains a smoother vertical mean profile, especially in the columns (marked by red dashline circles) distorted by stripe noise.

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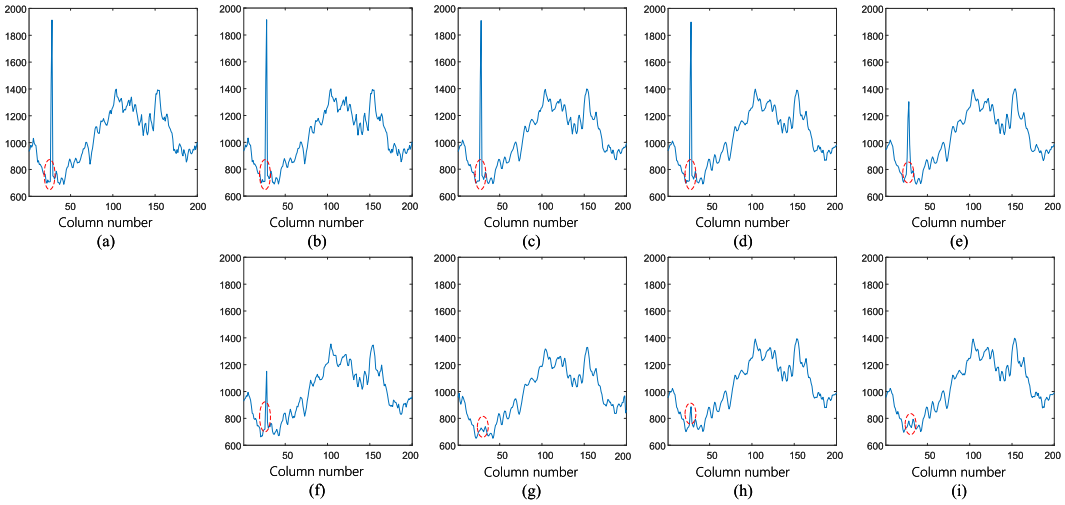


Fig. 14. Vertical mean profiles of band 128 in the ZY1E dataset (Subimage1). (a) Noisy. (b) LRTA. (c) BM4D. (d) LRMR. (e) HSID-CNN. (f) HSI-SDeCNN. (g) QRNN3D. (h) NG-Meet. (i) Proposed NL-3DCNN method.

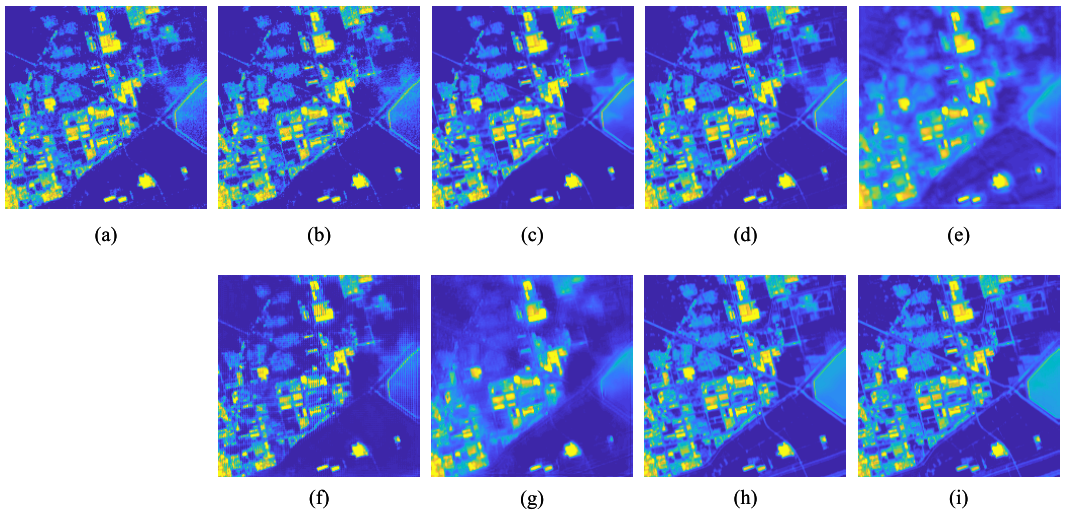


Fig. 15. (a) Original image with band 1 (Subimage2). (b) LRTA. (c) BM4D. (d) LRMR. (e) HSID-CNN. (f) HSI-SDeCNN. (g) QRNN3D. (h) NG-Meet. (i) Proposed NL-3DCNN method.

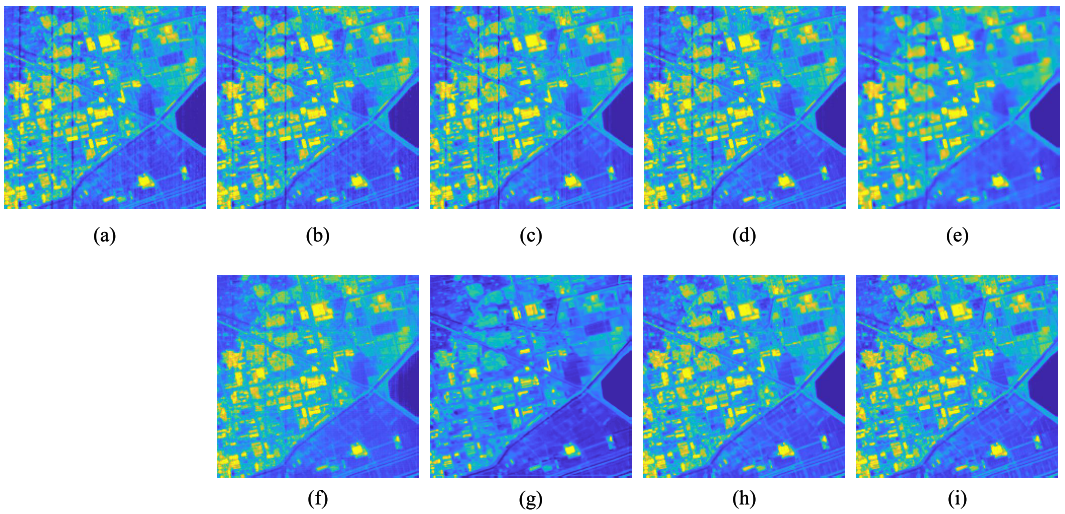


Fig. 16. (a) Original image with band 122 (Subimage2). (b) LRTA. (c) BM4D. (d) LRMR. (e) HSID-CNN. (f) HSI-SDeCNN. (g) QRNN3D. (h) NG-Meet. (i) Proposed NL-3DCNN method.

The second subarea was also clipped from the ZY1E dataset [shown in 7(c)]. The results in Figs. 15 and 16 show that LRTA, BM4D, and the LRMR cannot remove noise thoroughly. Meanwhile, NG-Meet and the proposed NL-3DCNN method can effectively remove Gaussian noise and slight stripe noise in the image. Furthermore, Fig. 17

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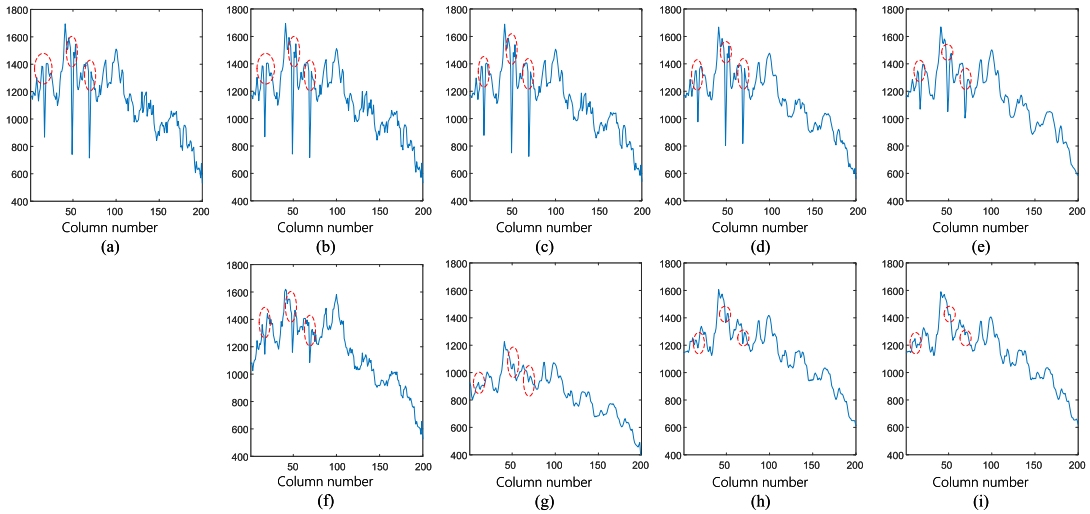


Fig. 17. Vertical mean profiles of band 122 in the ZY1E dataset (Subimage2). (a) Noisy. (b) LRTA. (c) BM4D. (d) LRMR. (e) HSID-CNN. (f) HSI-SDeCNN. (g) QRNN3D. (h) NG-Meet. (i) Proposed NL-3DCNN method.

demonstrates the corresponding vertical mean profiles of band 122 in the second subarea of ZY1E. It is clear that the NL-3DCNN performs smoother result than the NG-Meet method (marked with red dashline circles).

VI. Conclusion

In this article, a novel deep-learning-based nonlocal self-similarity denoising method, termed as NL-3DCNN, has been proposed for HSIs. The proposed method first represents an HSI in a low-dimensional subspace, and the corresponding representation coefficients are called eigenimages. Then a series of grouped similar 3-D patches are extracted from eigenimages and are fed to a deep network. Specifically, we train a 3-D CNN to fully explore the spatial information and the correlation of the eigenimages instead of the original HSIs. Our method yields better denoising results than the state-of-the-art methods both in the simulated and real data experiments.

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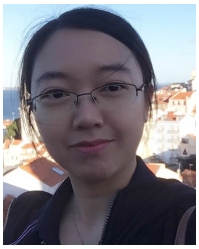
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